

Newton 法方程求根实验报告



代码:

```
#include <stdio.h>
#include <math.h>

double newton(double x, double y, double y0) {
    double x_1 = x - y / y0;
    return x_1;
}

int main() {
    double x_0 = 2;
    double epsilon = 0.5 * pow(10, -5);
    double f, f_d, x_1;
    int i = 1;
    int N = 20;

    while (i < N) {
        f = pow(x_0, 3) - 3 * x_0 - 1;
        f_d = 3 * pow(x_0, 2) - 3;

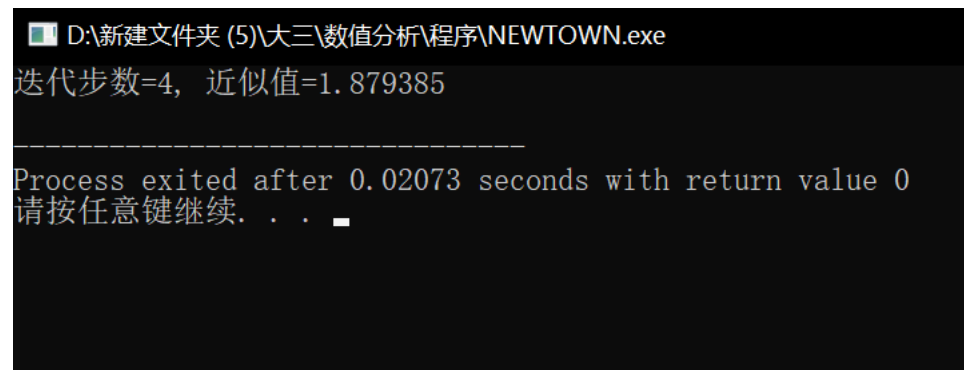
        x_1 = newton(x_0, f, f_d);
        if (fabs(x_1 - x_0) / (1 + fabs(x_1)) < epsilon) {
            printf("迭代步数=%d, 近似值=%f\n", i, x_1);
            break;
        }

        x_0 = x_1;
        i = i + 1;
    }

    if (i >= N)
        printf("超过最大迭代次数\n");

    return 0;
}
```

运行结果：



```
D:\新建文件夹 (5)\大三\数值分析\程序\NEWTOWN.exe
迭代步数=4, 近似值=1.879385
-----
Process exited after 0.02073 seconds with return value 0
请按任意键继续. . .
```